

Assistive Robot Capable of Drug Identification Based on Machine Vision

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Abstract. This study developed an assistive robotic vision system capable of identifying and sorting medicinal drugs packaged in trimmed blister packs. Utilizing a JetMax robotic arm integrated with an NVIDIA Jetson Nano and an HD wide-angle camera, the system identifies and delivers specific pills to designated locations. The YOLOv5 (You Only Look Once) algorithm was employed for real-time drug detection and recognition. A custom dataset comprising 20 medicine classes, each with 288 annotated images, totaling 5,760 images, was created and augmented through scaling, flipping, and contrast adjustments to mitigate model overfitting. The evaluation utilized a confusion matrix, with tests conducted over 600 real-time trials across medicine categories, including scenarios where no medicine was present. Results demonstrated an overall machine vision accuracy of 90.67%, with some classes achieving perfect accuracy. Mechanical performance, evaluated through robotic pick-and-place actions, achieved an 83.17% accuracy. Issues primarily arose from mechanical limitations related to the trimmed blister pack sizes, leading to some unsuccessful pick attempts despite correct drug identification. This study highlights both the efficacy of integrating YOLO-based machine vision with robotic systems for drug sorting and areas needing further refinement to enhance reliability and practical usability.

Keywords. Machine Vision, YOLO, Robotic Arm, Drug Identification, Assistive Robotics, Real-Time Detection, Jetson Nano.

1. Introduction

Machine Vision has become an essential component of robotic systems [1]. The vision system in an automated system captures and analyzes images to extract information such as depth information, color, geometry, and three-dimensional objects [2]. The Asia-Pacific region is witnessing substantial and rapid demographic changes. The Philippines, for instance, has a vast growth of the elderly and young population. The proportion of people aged 60 and older will increase from 6.7 percent in 2010 to 11.4 percent in 2030, while the group aged 0–14 years will make up 26.8% of the total population. As a result, the demand to improve health and social services increases and imposes considerable pressure on the country's limited medical resources [3]. Technologies such as Machine

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Vision have enhanced the feasibility of using robotics in people's daily lives, particularly in the healthcare system [4]. Most of these innovations productively affect people from all walks of life. In addition to improving humans' lives in the commonly known aspects, researchers today also focus on helping ameliorate the quality of life for individuals with functional diversity. Assistive Robots can take care of, perform domestic tasks, and assist in healthcare measures [5]. In addition, mobile robots are becoming a significant necessity for health-based assistance for the elderly [4].

Preliminary studies demonstrate that socially assistive robots deliver effective support in elderly care [5], yet only a limited number provide strong empirical evidence and even fewer examine mobile communicative platforms [6] so caregivers and patients predominantly rely on manual pill dispensing despite some error reduction via Automatic Tablet Dispensing Machines, which still permit filling inaccuracies and human - induced mistakes [7]; advances in three - dimensional fluorescence spectroscopy and deep-learning integration have enabled semi-quantitative drug identification in biofluids [8], but existing literature on blister-package pill recognition requires systematic revision to address real-world variability [9], and pharmaceutical manufacturers continue to investigate how material properties, production parameters, and tablet characteristics—such as size, weight, and particle-size distribution—influence final product quality [10]. Machine vision systems perform object detection by capturing optical signals and processing pixel-level data—brightness, color, and morphology—to execute predefined tasks, yet image distortions arising during acquisition, compression, or transmission degrade recognition performance [11]; visual information processing in mobile robotics encompasses image filtering, compression, and three-dimensional perception [12], and optimizing camera exposure enhances tasks such as optical character recognition [13], while morphological analysis and separation into multiple color spaces simplify feature extraction [14] and digital image processing improves production efficiency, elevates product quality, and mitigates human-inspection errors [15]. Image-processing frameworks and computer-vision classifiers successfully identify defects in agricultural products such as Fresh Excelsa Beans [16] and tomato ripeness and size [17]—often by coupling traditional techniques with machine learning to boost performance [18]—and emerging YOLO variants extend these applications across AI domains [19], though they demand extensive experimental validation to confirm solution viability [20], and in biomedical contexts, image processing supports systems ranging from dental-bracing information tools [21] to life-buoy detection [22]. Robotic manipulators—mechanical arms driven by rotating actuators—have achieved high positional accuracy in industrial and healthcare environments [23][24], and object-detection algorithms, typically implemented in Python, underpin real-time applications from fish-species selection [25] to biometric tongue-print identification [26], with deep-learning detectors such as YOLO and RetinaNet delivering reliable performance for pharmaceutical object models and surpassing alternatives like SSD in speed and accuracy [27][28]; nevertheless, integrating machine vision with mobile robotic platforms remains cost- and complexity-intensive [29], and today's systems require enhanced flexibility, efficiency, and user-centric features to realize truly assistive care robots [30].

With all the technological advancements in Machine Vision and Robotics, designing economic systems that help humans and augment their abilities to do certain tasks has become necessary. However, most effects have only focused on artificial application scenarios that do not entail user feedback integration or users' intentions and goals. Consequently, they tend to be of limited use when it is necessary to assist humans. Technologies such as assistive robots should also consider an operational paradigm in

which the user is central, such that the user can influence and be influenced by the system. The available robotic systems integrated with machine vision need more flexibility. The present systems can filter according to limited parameters only; Furthermore, most systems use preset joint angles to set the position of the robotic arm. This further reduces the system's flexibility, where the object's position to be picked up and placed must be exact. Moreover, the present drug identification models implemented using the YOLO algorithm only focus on pills' imprinted characters, shapes, colors, and whole blister packages for identification.

Generally, this study aims to utilize the emerging role of machine vision in human-machine interaction. This study will develop a robotic vision system as a user-friendly device to sort a specific drug in a trimmed blister package in a designated location. Specifically, it will focus on (1) utilizing the JetMax robotic arm and its built-in HD wide-angle camera with Jetson NANO to deliver an identified pill to a designated location, (2) using the YOLO (You Only Look Once) algorithm to detect and recognize the drug in a trimmed blister package, and (3) evaluating the machine vision system through a confusion matrix.

People's lives in the digital age have been consistently more accommodating as technological advancements rapidly arise. Robotics and artificial intelligence advancements have allowed social assistive robots to improve their potential to assist humans in various conditions. Every individual will soon own a robot that will imperatively guide humans in everyday tasks in today's digital age. A new paradigm in nursing care involving Assistive Robots is being investigated. This study will significantly add to the application of technology that helps humans in their daily lives. This study will provide an exciting opportunity to advance human knowledge of human-machine interaction in the biomedical field. This study will offer new insights into machine vision in assisting humans, specifically the elderly.

This study is limited to detecting and recognizing trimmed blister packages of medicinal drugs. The proponents will create a dataset for the study comprising the images of twenty (20) medicinal drugs for pill identification, namely Amlothix, Rosuvastatin, Avator, Ascof-Forte, Cetirizine, Neozep, Poten-Cee, Biogesic, C-Lium Fibre, Buscopan, Bioflu, Xtracee, Advil, Fern-C, Vit-Eye, Mefenamic acid, Loviscol, Mucosolvan, Glycemet, and Decolgen-Forte. It is also limited to picking up and delivering the chosen pill one at a time. The system is limited to Jetson Nano, JetMax for the robotic arm, an HD wide-angle camera, and the YOLO (You Only Look Once) Algorithm. It is beyond the scope of this study to utilize machine vision in fields other than biomedical and human-machine interaction. This study will not cover identifying medicinal drugs in other packaging types.

2. Methodology

2.1. Conceptual Framework

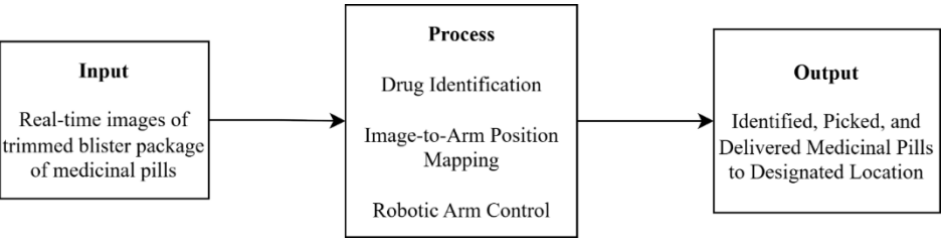


Figure 1. Conceptual Framework

Figure 1 displays the conceptual framework of the study. The input consists of real-time images of trimmed blister packages of medicinal pills, which are captured and analyzed by the system. The process involves three main stages: drug identification, image-to-arm position mapping, and robotic arm control. First, drug identification is performed using an object detection algorithm to recognize specific types of medicines. Next, the identified pill's location in the image is translated into real-world coordinates that the robotic arm can understand. Finally, the robotic arm is commanded to pick up and place the identified medicine into its designated location. The output of this system is the identification, picking, and delivery of medicinal pills to their respective sorting areas.

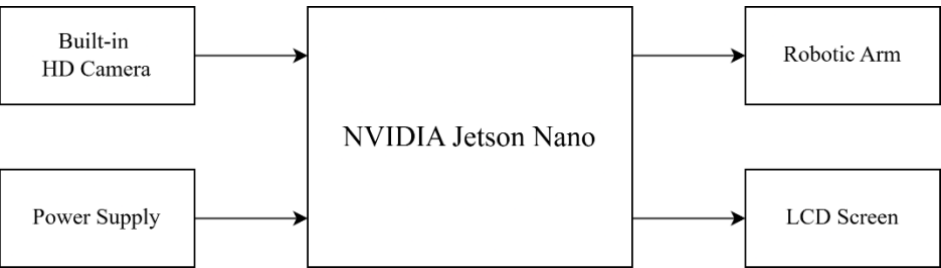


Figure 2. Hardware Block Diagram

Figure 2 shows the hardware design of the system, which includes the NVIDIA Jetson Nano, built-in HD camera, robotic arm, power supply, and LCD screen. The Jetson Nano, running Ubuntu 18.04, is equipped with an ARMv8 Processor, 4 GB of memory, and an NVIDIA Tegra X1 GPU with CUDA 10.2, allowing real-time object detection and control. The system uses Python 3.8 as its programming language and PyTorch 1.8 for implementing the YOLOv5 model. The camera captures images of trimmed blister packages, which are analyzed by the Jetson Nano to identify medicines. The robotic arm picks and sorts the pills, while the LCD screen displays real-time outputs.

2.2. System Flowchart

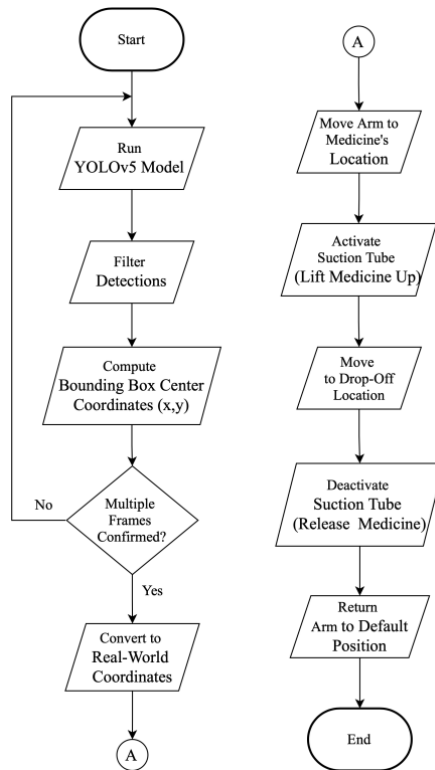


Figure 3. Main Flowchart

Figure 3 illustrates the process of identifying, picking, and sorting medicinal pills from trimmed blister packages using an automated system guided by a JetMax Robotic Arm. The process begins with running the YOLOv5 model to detect and classify the medicine present in the captured image. After detection, the system filters out low-confidence detections to ensure that only accurate identifications are processed. Once filtered, the system computes the center coordinates (x, y) of the bounding box around the detected medicine to determine its precise location in the image.

To ensure that the detection is stable and not an error, the system performs a multiple-frame confirmation, checking whether the same medicine appears consistently over several frames. If confirmed, the identified coordinates are converted into real-world coordinates, which the robotic arm can use for accurate positioning in its workspace. After determining the real-world location of the medicine, the robotic arm moves to the position of the identified pill. The system then activates the suction tube to lift the medicine securely. Once lifted, the arm moves to a designated drop-off location where the pill is to be sorted. Upon reaching the drop-off point, the system deactivates the suction tube to release the medicine carefully into the assigned container. Finally, the robotic arm returns to its default position, ready to repeat the cycle for the next pill detected in the trimmed blister package.

2.3. Experimental Set-Up

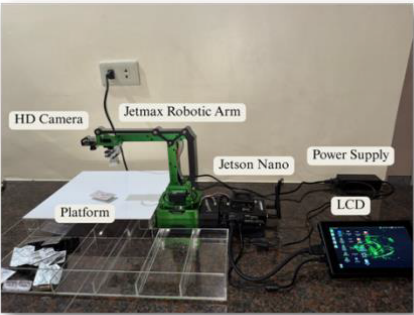


Figure 4. Experimental Set-Up

Figure 4 illustrates the experimental setup used for the automated sorting of trimmed blister-pack medicines. The system integrates a JetMax robotic arm equipped with a suction-based gripper, positioned on a shared platform with an NVIDIA Jetson Nano. This controller runs a YOLOv5 object detection model in real time using a live camera feed. Each medicine blister is placed on a flat white surface, where the camera captures the scene, the model generates labeled bounding boxes, and the coordinates are extracted by the control software. The robotic arm then aligns with the identified object, picks it up using suction, and transfers it to the appropriate drop-off location. A touchscreen interface displays live feedback, including detection outputs and operational status. The sorting layout consists of four labeled containers, each corresponding to a distinct pharmaceutical category. These were predefined according to standard classifications for logical medication grouping. Box 1 handled cardiovascular and metabolic drugs (Amlothix, Rosucol, Avator, Glycemet), Box 2 managed pain and fever relief (Biogesic, Mefenamic Acid, Advil, Buscopan, Bioflu), Box 3 was designated for cough, cold, and allergy treatments (Ascof Forte, Loviscol, Mucosolvan, Neozepe, Decolgen Forte, Askey), and Box 4 contained vitamins and dietary supplements (Poten-Cee, Fern-C, Xtracee, Vit-Eye, C-Lium Fibre). Once classification is complete, the robotic arm executes the corresponding sorting action by placing each medicine into its assigned box.

Table 1. Software and Hardware Setup

Name	Experimental Setup
Epochs	100
Batch Size	4
Image Size	160 x160
Pre-trained Model	yolov5s.pt
TensorRT	last.trt
Programming Language	Python 3.8
Deep Learning Framework	PyTorch 1.8
CPU	ARMv8 Processor
Memory	4 GB
GPU	NVIDIA Tegra X1
CUDA	10.2
Software Platform	Ubuntu 18.04

Table 1 presents the experimental setup detailing software and hardware configurations used in training, validating, and deploying the robotic vision system. Software parameters specify training iterations set at 100 epochs to ensure stable model convergence, a batch size of 4 selected to fit the 4 GB memory constraint of the NVIDIA Jetson Nano, and an image size of 160×160 pixels chosen to balance inference accuracy and computational efficiency. Additionally, researchers optimized the YOLOv5s pre-trained model (yolov5s.pt) into a TensorRT engine (last.trt), enabling efficient, real-time object detection. Hardware specifications include Python 3.8 and PyTorch 1.8 as the programming environment and deep-learning framework, respectively, alongside an ARMv8 processor, NVIDIA Tegra X1 GPU, CUDA 10.2 for GPU acceleration, and Ubuntu 18.04 as the operating system. Together, these configurations ensure compatibility, replicability, and effective operational deployment of the robotic vision system.

2.4. Custom Dataset of the Trimmed Blister Medicinal Packages - Back Model

The dataset used in this study is custom-made, consisting of 20 classes with 288 images of the back part per class, totaling 5,760 images. Images were annotated by their class name using Labelling software. To address the limited sample size and prevent model overfitting, offline augmentation methods—including scaling, flipping, and contrast adjustments—were applied. These techniques expanded the dataset, ensuring sufficient diversity and consistency, and minimizing interference from variations in the sample space. During training, the dataset was split into training and validation sets at an 80% to 20% ratio. Examples of images of trimmed blister medicinal packages are presented in Figure 5.



Figure 5. Sample Images of Trimmed Blister Medicinal Packages

3. Results and Discussion

3.1. Machine Vision System

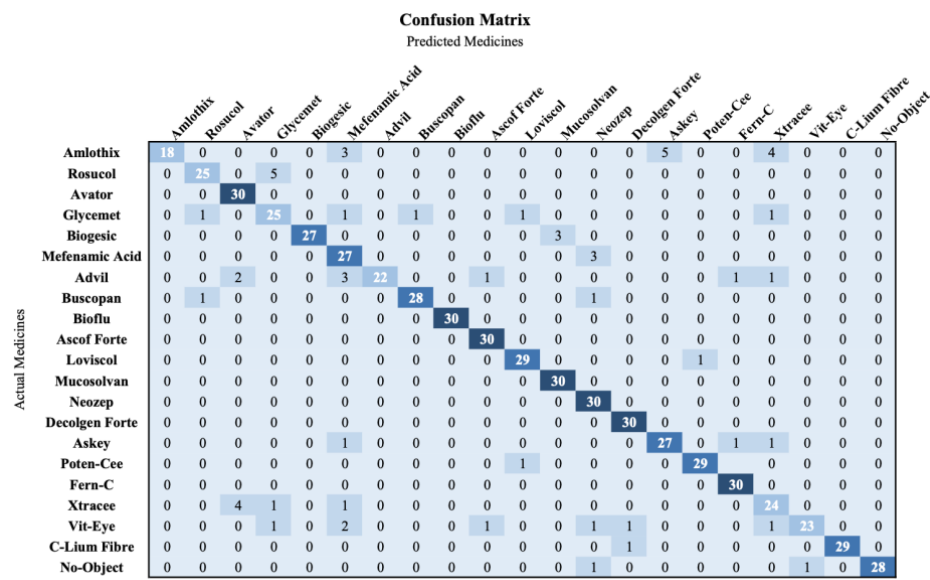


Figure 6. Confusion Matrix of the Machine Vision System using Real-Time Images of Medicines

For the testing procedure of the machine vision system, 30 trials of real-time image classification were conducted per medicine class, including a “No-Object” category, for a total of 630 trials. The multi-class confusion matrix for this test dataset is presented in Figure 6. The medicine classes Bioflu, Ascof Forte, Mucosolvan, Decolgen Forte, Fern-C, and Avator demonstrated perfect accuracy, with all 30 samples correctly classified. Other medicines exhibited minor inaccuracies; Rosucol and Glycemet each had 5 misclassified samples, while Amlothix had 12 misclassifications, primarily confused with Xtracee and Mefenamic Acid. Additionally, Xtracee showed 6 misclassifications, mostly confused with Amlothix. The “No-Object” category successfully achieved 28 correct identifications out of 30 trials, indicating the system’s robust capability to recognize scenarios where no medicine was present. Overall, these results suggest that the developed machine vision model performs with high accuracy, highlighting specific classes that may benefit from additional refinement to improve system reliability and effectiveness.

Table 2. Accuracy of the Machine Vision System

Medicine Class	Number of Trials	Correct Prediction	Accuracy Rate
Amlothix	30	18	60%
Rosucol	30	25	83.33%
Avator	30	30	100%
Glycemet	30	25	83.33%
Biogesic	30	27	90%
Mefenamic Acid	30	27	90%
Advil	30	22	73.33%
Buscopan	30	28	93.33%
Bioflu	30	30	100%
Ascof Forte	30	30	100%
Loviscol	30	29	96.67%
Mucosolvan	30	30	100%
Neozep	30	30	100%
Decolgen Forte	30	30	100%
Askey	30	27	90%
Poten-Cee	30	29	96.67%
Fern-C	30	30	100%
Xtracee	30	24	80%
Vit-Eye	30	23	76.67%
C-Lium Fibre	30	29	96.67%
No Object	30	28	93.33%
Overall Accuracy			90.67%

Table 2 illustrates the accuracy results of the machine vision system across 20 medicine classes; each was tested through 30 trials. Classes such as Avator, Bioflu, Ascof Forte, Mucosolvan, Neozep, Decolgen Forte, and Fern-C achieved perfect accuracy rates (100%), reflecting the model’s robust ability to distinguish these medicines. However, the system showed lower accuracy rates with certain medicines, particularly Amlothix (60%), Vit-Eye (76.67%), and Advil (73.33%), indicating notable recognition challenges. These lower accuracy results may suggest visual similarities with other medicines, requiring further refinement of the training dataset or algorithmic adjustments. The overall accuracy achieved by the system is 90.67%, highlighting strong general performance.

3.2. Robotic System Integrated with the Machine Vision System

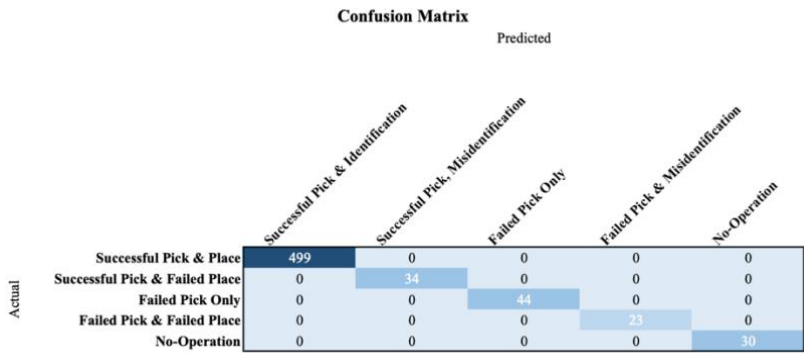


Figure 7. Confusion Matrix of the Robotic System Integrated with the Machine Vision System

Figure 7 summarizes the combined performance of the robotic pick-and-place system and the machine vision classification subsystem. The diagonal values represent correct predictions, indicating perfect alignment between predicted and actual outcomes. The system achieved 499 fully successful trials, reflecting accurate identification coupled with successful robotic picking and placing. However, the matrix also reveals certain mechanical and identification challenges. Notably, 34 trials were correctly picked but misidentified, suggesting minor inaccuracies within the vision subsystem. Additionally, mechanical issues were evident, with 44 instances of failed picking attempts and 23 trials where both picking and placing failed, despite correct identification. The "No-Operation" category showed flawless performance with 30 correct trials, confirming the system's effectiveness in accurately identifying scenarios requiring no action.

Table 3. Accuracy of the Robotic System Integrated with the Machine Vision System

Action Done	Number of Trials	Correct Prediction	Accuracy Rate
Successful Pick and Identification	600	499	83.17%
Successful Pick, Misidentification	600	34	5.67%
Failed Pick, Successful Identification	600	44	7.33%
Failed Pick, Failed Identification	600	23	3.83%
No Operation	600	30	5%

Table 3 summarizes the integrated performance of the robotic pick-and-place system with the machine vision identification subsystem. The "Successful Pick and Identification" category recorded an accuracy of 83.17%, with 499 correct predictions out of 600 trials. This highlights strong overall system integration, demonstrating effective identification paired with precise robotic execution. However, there were cases categorized as "Failed Pick, Successful Identification," with 44 occurrences (7.33%). These instances occurred due to difficulties associated with picking up medicines from trimmed blister packages, indicating that mechanical issues, primarily due to blister size and shape, hindered successful robotic grasping despite accurate vision identification.

4. Conclusion and Future Works

Overall, the objectives of this study were successfully met by developing a robotic vision system capable of accurately identifying and sorting medicinal drugs packaged in trimmed blister packs. The integration of a JetMax robotic arm with an NVIDIA Jetson Nano and an HD wide-angle camera enabled precise drug identification and sorting. Employing the YOLO (You Only Look Once) algorithm demonstrated high efficiency and accuracy, achieving an overall machine vision accuracy of 90.67%. The robotic pick-and-place subsystem showed effective performance, attaining an accuracy rate of 83.17%. However, mechanical challenges arose primarily due to the varying dimensions and characteristics of trimmed blister packages, causing occasional pick-up failures despite accurate visual detection. The integrated robotic vision approach significantly contributes to automation in pharmaceutical sorting and dispensing tasks, demonstrating clear potential for practical healthcare applications.

To further enhance the reliability and real-world applicability of the system, several avenues for future research are suggested. First, expanding and diversifying the dataset to encompass a broader range of lighting conditions, occlusions, and object variations will improve model robustness. Second, experimenting with more sophisticated deep learning models or hybrid methods could lead to higher precision in drug identification. Finally, future studies should incorporate evaluations in clinical settings to identify deployment challenges and refine the system for practical healthcare use. These improvements would significantly strengthen the effectiveness and integration of robotic vision systems in medical environments.

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